

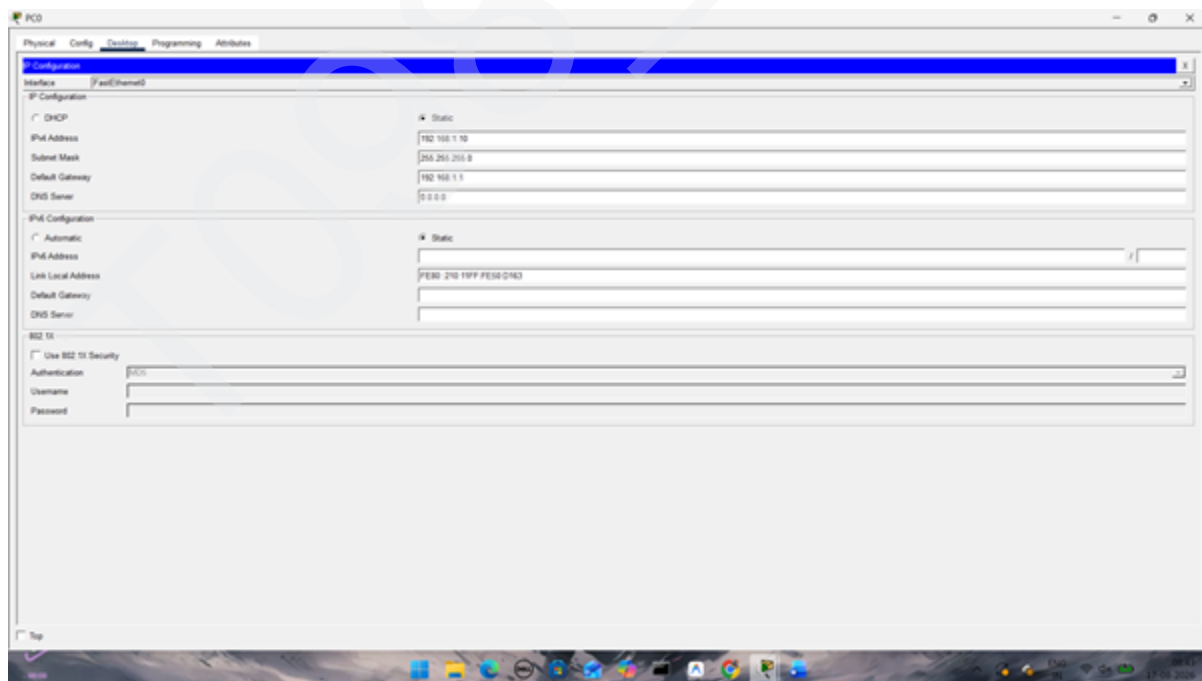
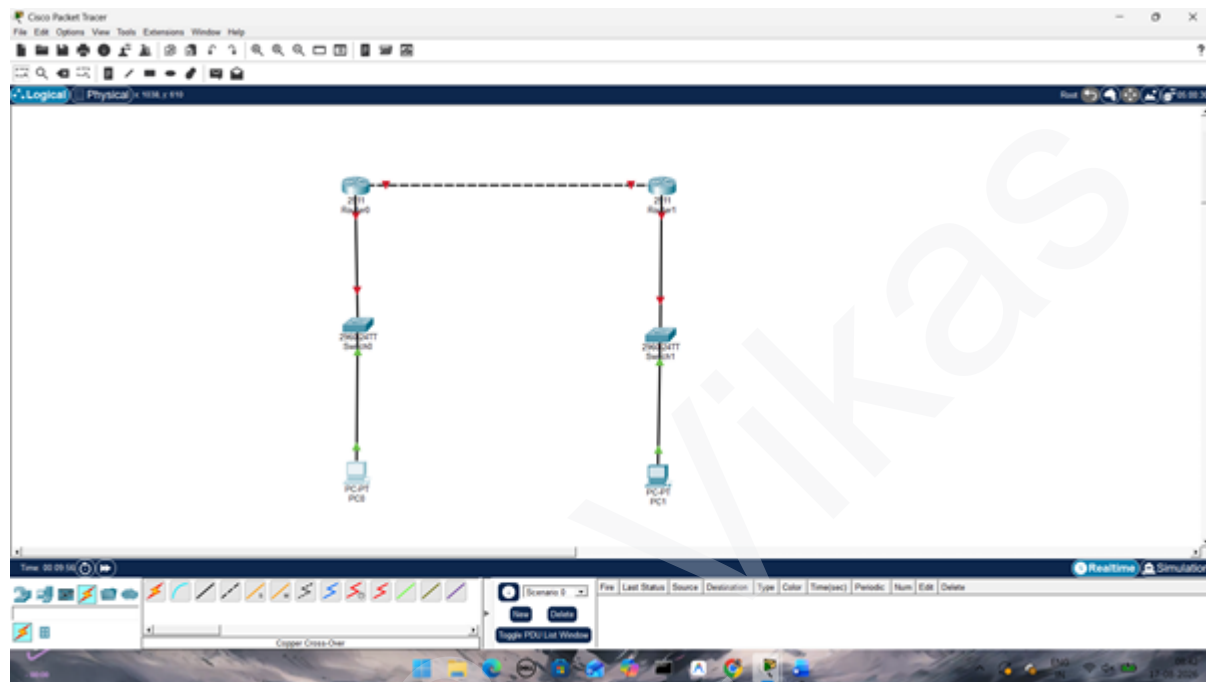
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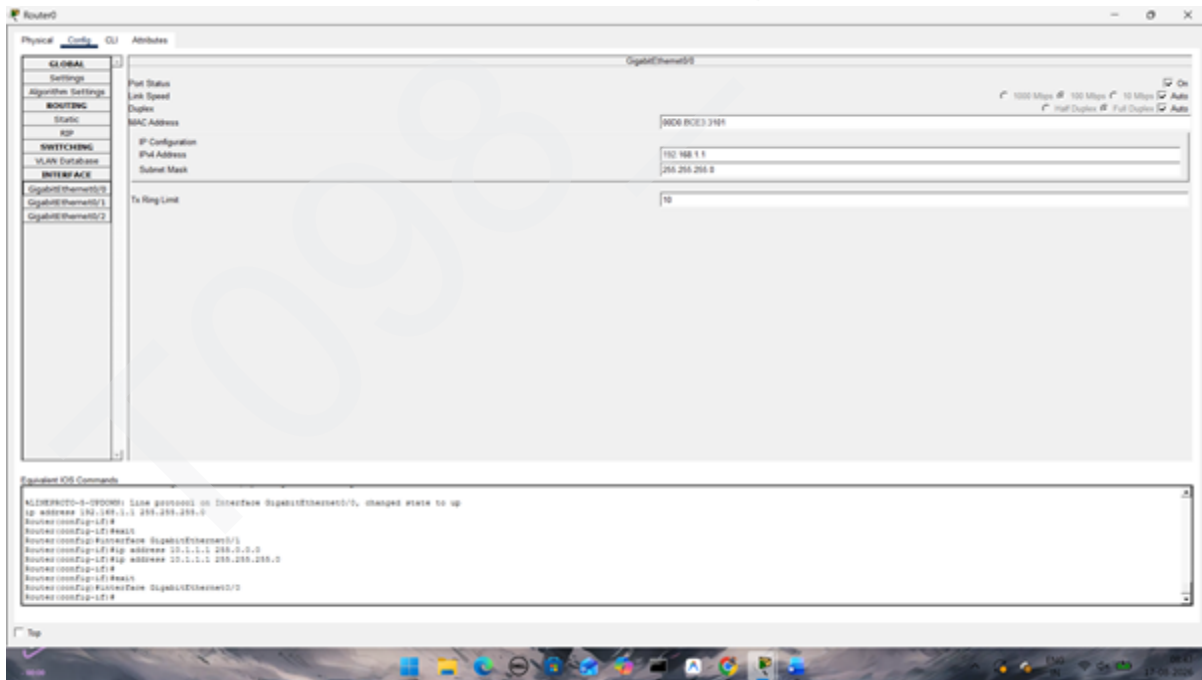
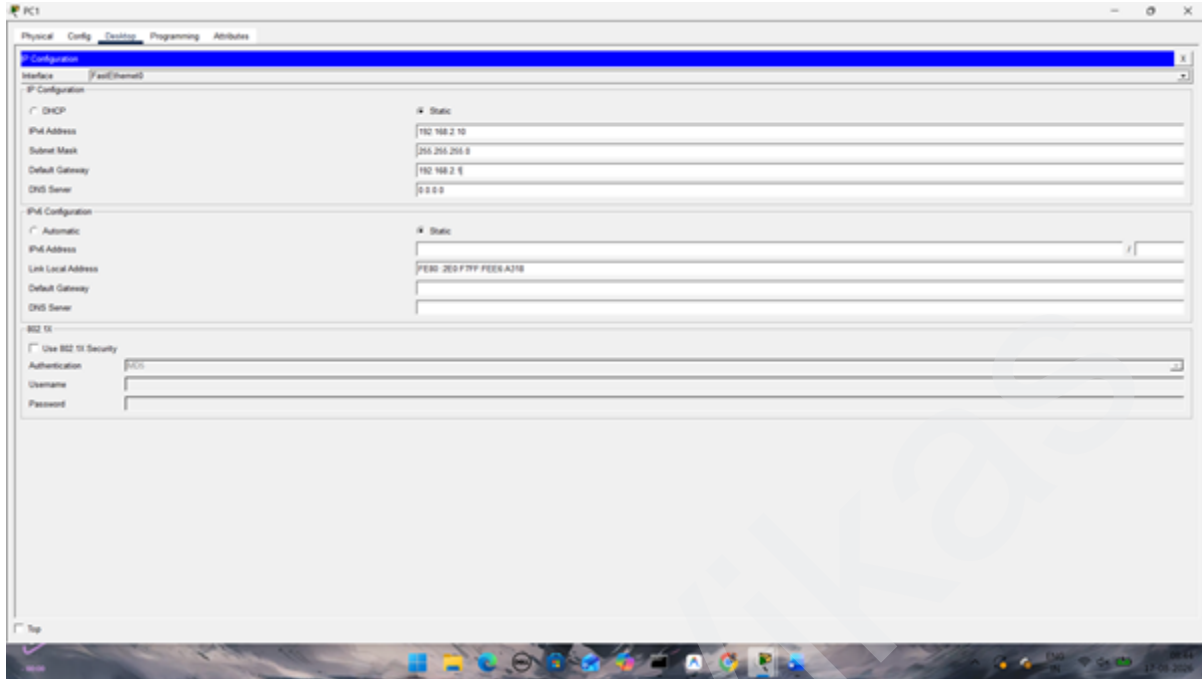
PRACTICAL NO 6

AIM: Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.

1. Network Topology & Devices



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IP Address Configuration

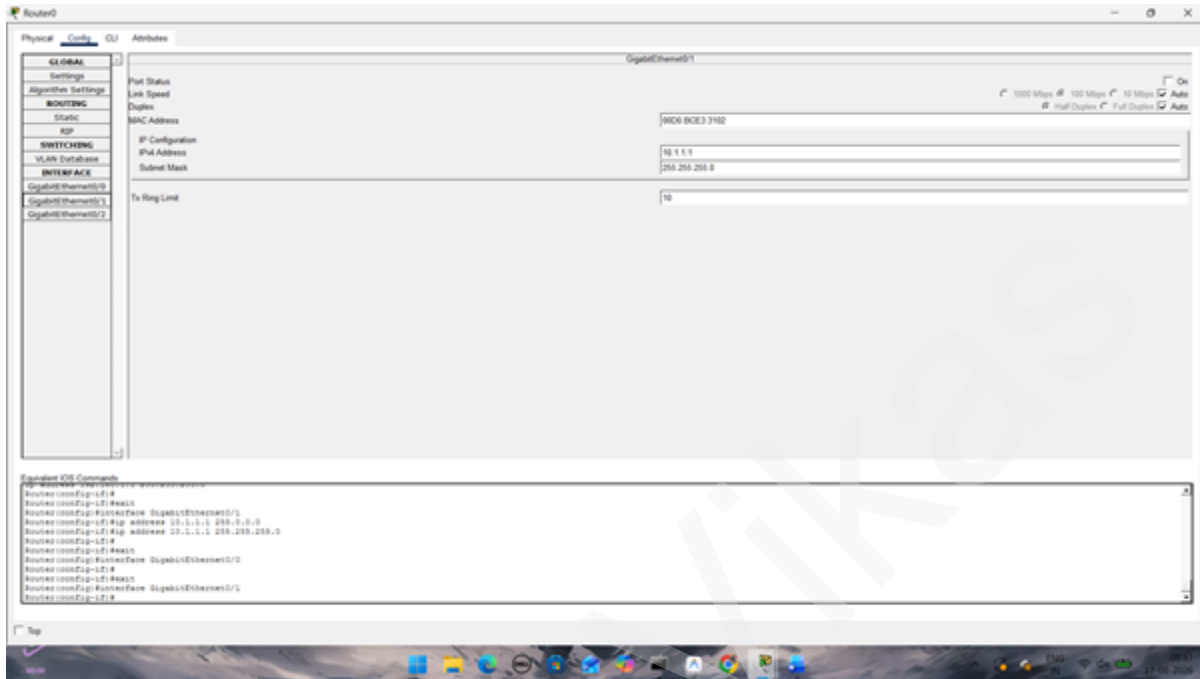
- **Site 1 (Left):**

Vikas Pal
T098

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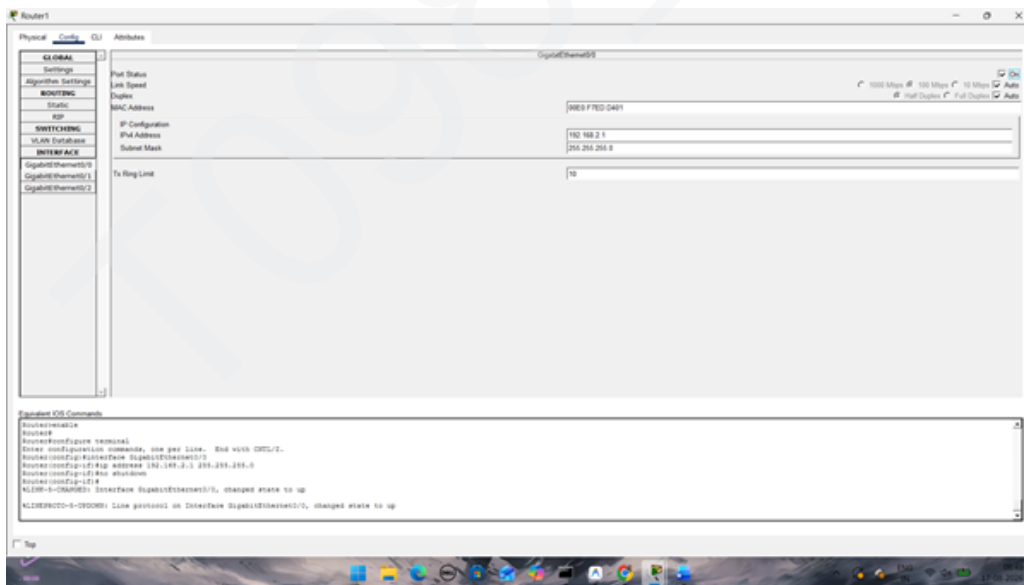
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- **Router0:** LAN (Gig0/0) = 192.168.1.1 | WAN (Gig0/1) = 10.1.1.1
- **PC0:** IP = 192.168.1.10 | Gateway = 192.168.1.1



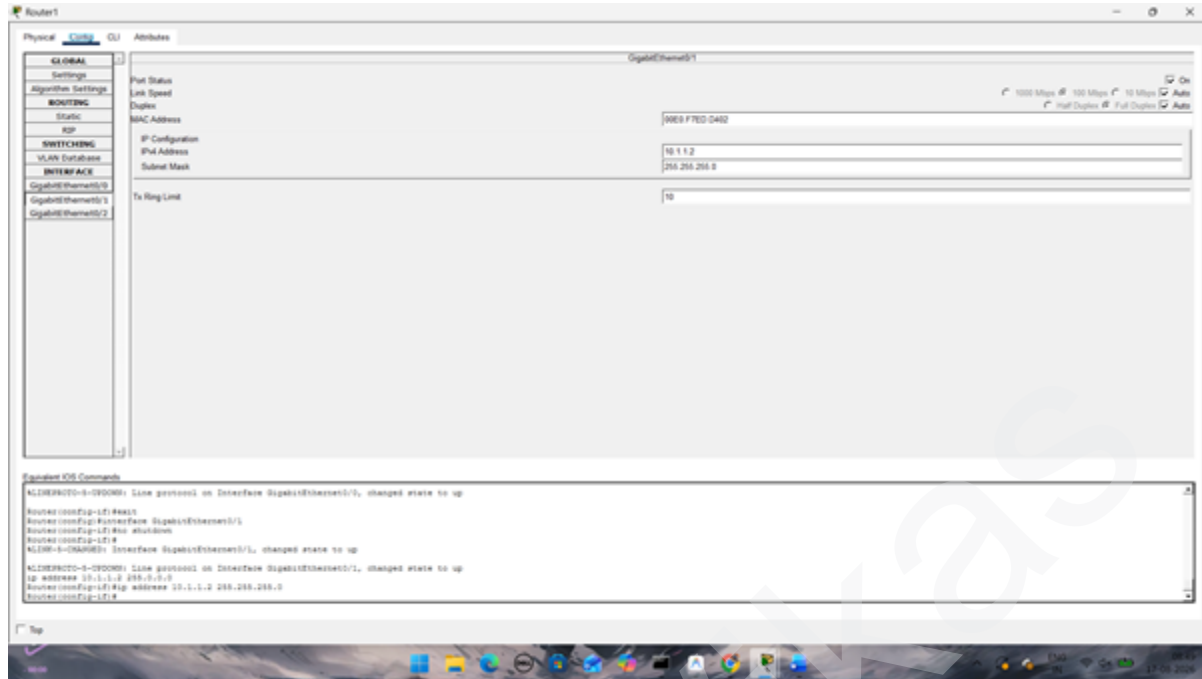
Site 2 (Right):

- **Router1:** LAN (Gig0/0) = 192.168.2.1 | WAN (Gig0/1) = 10.1.1.2
- **PC1:** IP = 192.168.2.10 | Gateway = 192.168.2.1



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IPsec Phase 1 (ISAKMP)

- Created ISAKMP policy 10 on both routers with AES encryption, SHA hashing, and pre-shared key authentication.
- Set the shared secret key: crypto isakmp key vpnsecret123 address [Peer-WAN-IP].

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IPsec Phase 2 (Transform Set & ACL)

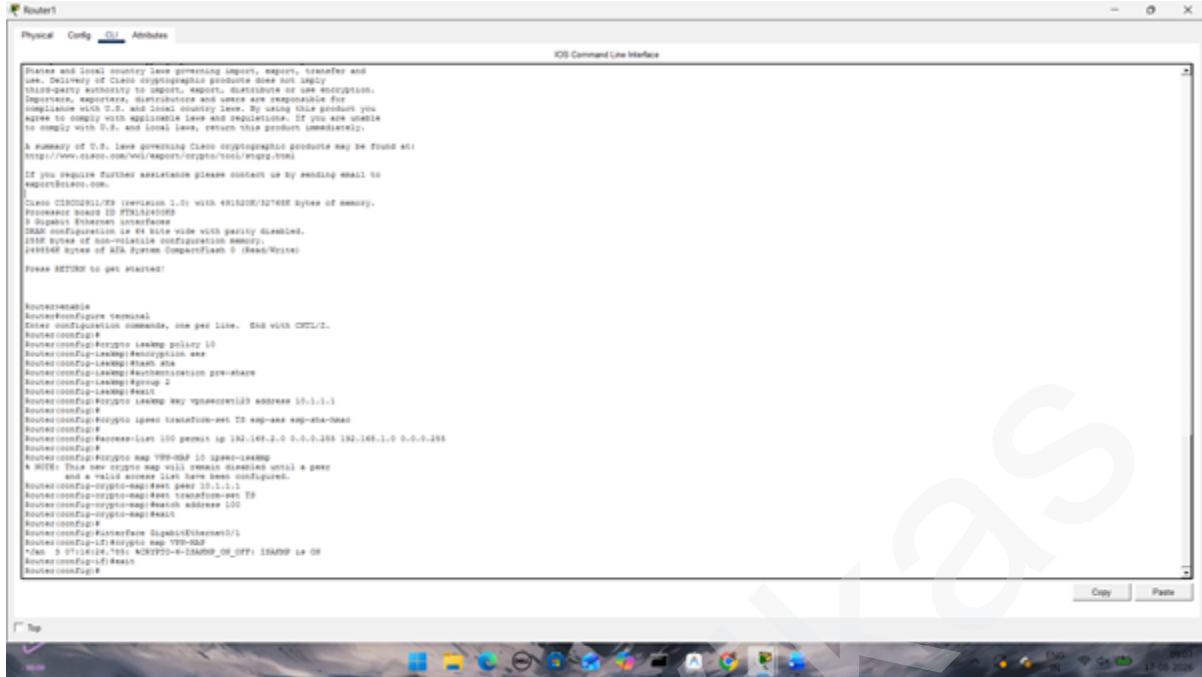
- Created a transform set named **TS** using esp-aes and esp-sha-hmac.
- Created Extended Access Control List (ACL) **100** to permit and identify interesting traffic moving between 192.168.1.0 and 192.168.2.0.

Crypto Map Application

- Created a crypto map named **VPN-MAP** that linked the peer IP, the transform set, and the ACL.
- Applied the crypto map to the GigabitEthernet0/1 (WAN) interface on both routers to arm the VPN.

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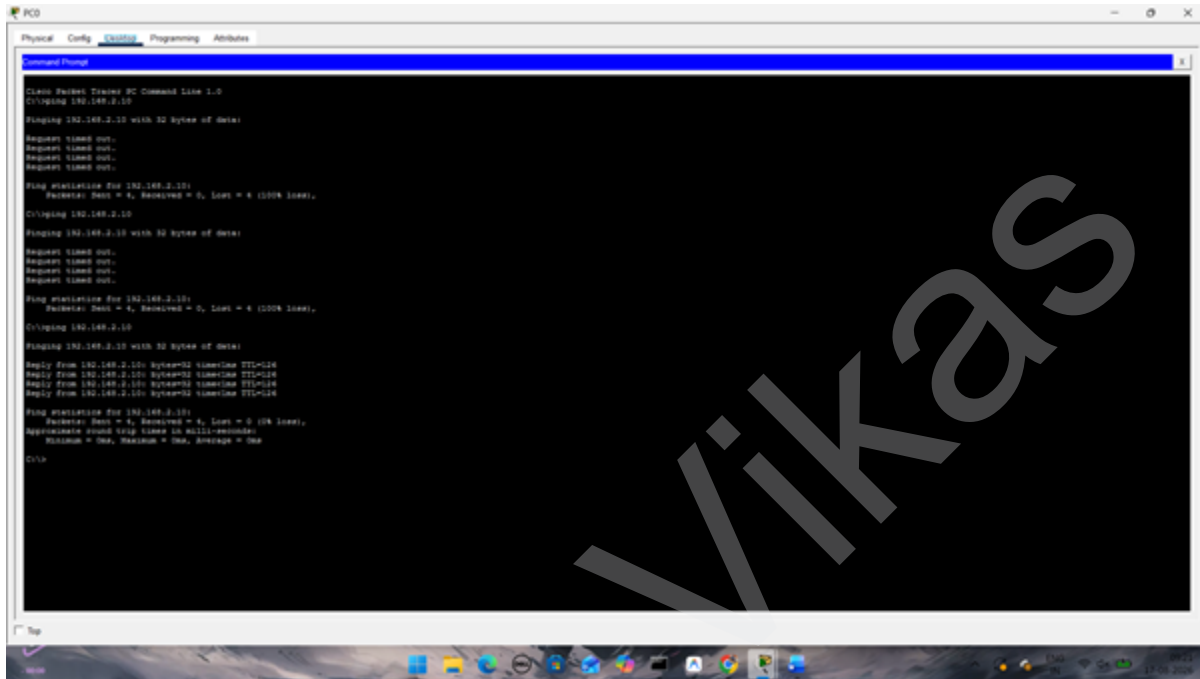


Verification and Output

- **Triggering the Tunnel:** Initiated an ICMP ping from PC0 (192.168.1.10) to PC1 (192.168.2.10). The first packets timed out during negotiation, followed by successful replies, indicating the tunnel was established.
- **Verifying Encryption:** Executed the `show crypto ipsec sa` command on the routers. The `pkts encaps` (encapsulated) and `pkts decaps` (decapsulated) counters successfully incremented, proving the data crossing the WAN link was actively encrypted.
- **Result:** The Site-to-Site IPsec VPN was successfully deployed and verified.

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The screenshot shows a Windows PC desktop with a taskbar at the bottom. A Command Prompt window is open, displaying the output of a 'C:\>ping 192.168.2.10' command. The output shows three successful pings with 0% loss and round-trip times of 0ms. A large, semi-transparent watermark 'Vikas' is overlaid diagonally across the center of the image.

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packet: Sent = 4, Received = 0, Loss = 4 (100% loss),
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

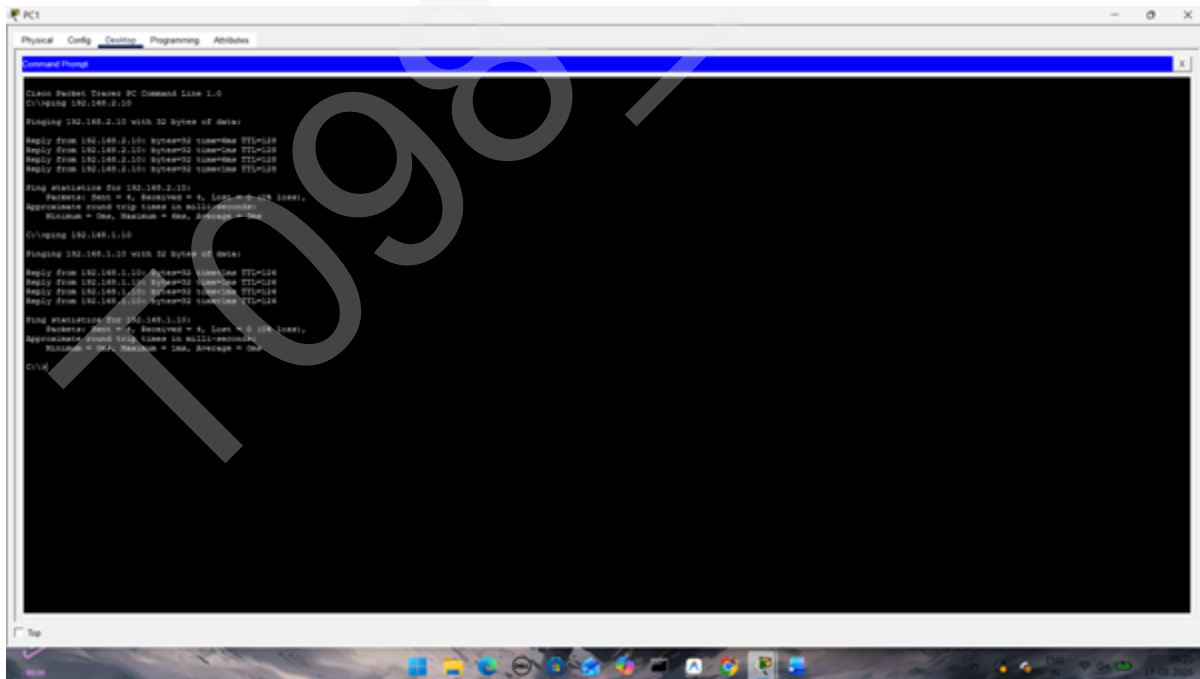
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.2.10:
    Packet: Sent = 4, Received = 0, Loss = 4 (100% loss),
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128
Reply from 192.168.2.10: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.2.10:
    Packet: Sent = 4, Received = 4, Loss = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```



The screenshot shows a Windows PC desktop with a taskbar at the bottom. A Command Prompt window is open, displaying the output of a 'C:\>ping 192.168.1.10' command. The output shows three successful pings with 0% loss and round-trip times of 0ms. A large, semi-transparent watermark 'Vikas' is overlaid diagonally across the center of the image.

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=0ms TTL=128
Reply from 192.168.1.10: bytes=32 time=0ms TTL=128
Reply from 192.168.1.10: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.10:
    Packet: Sent = 4, Received = 4, Loss = 0 (0% loss),
    Approximate round trip times in milli-seconds:
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Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=0ms TTL=128
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Ping statistics for 192.168.1.10:
    Packet: Sent = 4, Received = 4, Loss = 0 (0% loss),
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C:\>
```